

Mohamed Sami GHRAB



Lausanne, Switzerland | ghrab3355@gmail.com | +41 78 334 91 79
[linkedin.com/in/mohamed-sami-ghrab](https://www.linkedin.com/in/mohamed-sami-ghrab)
Languages: English, French, Arabic, German (basic)

EDUCATION

EPFL, Swiss Federal Institute of Technology Lausanne

Master in Data science ongoing
Bachelor in Communication Systems 2024

AREAS OF EXPERTISE

- Data Visualization
- Computer Security
- Machine Learning
- Large-scale Data
- Database Systems
- Foundation models & GenAi
- Natural Language Processing
- Optimization for Machine Learning
- Reinforcement Learning

Skills : Python (Pandas NumPy Statsmodel Matplotlib Seaborn Scikit-learn PyTorch PySpark NLTK) SQL Scala Java C R

PROJECTS

Data Story Project — “When Anger Speaks Fluently” (CS-401 — Applied Data Analysis) jan2026
Development of an interactive data story analyzing 1M+ Reddit interactions to study language style, sentiment and community behavior . Website: [website-ada](#)

- Developed and deployed the website using React, TypeScript, Jekyll, Tailwind CSS, and GitHub Pages
- Integrated interactive visualizations including PCA clustering, sentiment analysis, and subreddit interaction networks

Fine-tuned Aerial Image Segmentation (ALS) dec2025

Developed and evaluated deep learning models for aerial image semantic segmentation using Python and PyTorch.

- Implemented and fine-tuned U-Net (ResNet18), SegFormer (mit_b2), DeepLab, and diffusion-based segmentation models.
- Built data preprocessing and augmentation pipelines with OpenCV and Albumentations, and optimized performance using BCE + Dice Loss, F1 Score, and Dice metrics.

Picky — AI Emotional Recommendation Platform (Team Project) dec 2025
Development of an emotion-aware recommendation Telegram bot using LLMs and Supabase

- Designed and implemented an LLM-powered emotion detection pipeline (18 emotion categories) using the OpenRouter API
- Built a Telegram bot recommending movies, music, and books based on users’ emotional states
- Created analytics dashboards in Grafana to track satisfaction, retention, decision time, and emotional-fit KPIs

Predictive Modeling for Early Heart Attack Detection nov 2024
Estimation of heart attack outcomes from patient behavioral data, enhancing early detection and supporting medical decision-making

- Developed a regularized logistic regression model from scratch using NumPy on 320,000 records with 350 features from the BRFSS 2015 dataset to predict heart attack risk
- Conducted data preprocessing, including cleaning, handling missing values, standardizing, balancing, and polynomial expansion to enhance model reliability
- Achieved high F1 and accuracy scores via hyperparameter tuning using cross-validation; ranked top 5% on AICrowd

PROFESSIONAL EXPERIENCE

Research Engineer – Clinical LLM Fine-tuning & Representation Analysis (LiGHT Lab) feb2026-ongoing

- Fine-tuned Qwen3-32B on clinical Q&A data using Supervised Fine-Tuning (SFT) and analyzed the effect of subsequent preference tuning (DPO / Offset DPO) on internal model representations
- Built an end-to-end hidden state extraction and linear probing pipeline across 8 model checkpoints on a 4x A100 HPC cluster.
- Ran cosine similarity, PCA geometry, and logistic probe classifiers on paired chosen/rejected representations across 8 transformer layers.
- Key finding: preference tuning consistently improves clinical output quality (~+13% Jaccard over base) without reorganizing internal latent geometry